

Lecture Notes no 4

Transformations on the Response Variable when the Assumptions of Homogeneity of Variances is Violated.

One common cause for heterogeneity of variances between levels of the treatment factor is a non linear relationship between the response and the treatments. One approach to correct the situation is to transform the response variable.

One way to recognize the need for a variance stabilizing transformation is to examine the plot of the residuals (or the jackknifed residuals) vs the fitted values of the response variable. Based on that plot the following transformations are suggested:

Relationship between $\sigma^2 = \text{Var}(\epsilon_{ij})$ and μ_i	Transformation
$\sigma \propto \mu_i^2$	$1/Y$
$\sigma \propto \mu_i^{3/2}$	$1/Y$ or $\sqrt{Y}$
$\sigma \propto \mu_i$	$\log(Y)$
$\sigma \propto \mu_i^{1/2}$	$\sqrt{Y}$

Distribution Based Transformations

In some data sets it will be clear that the distribution of the errors and therefore of the response variable is most likely not normals. The following transformations can be employed if it is acceptable to do so.

Response Distribution-Based Transformations

Response Distribution	Transformation
Binomial	$\sin^{-1} \sqrt{Y/n}$
Poisson	$\sqrt{Y}$ or $\sqrt{Y + 1/2}$
Lognormal	$\log(Y)$

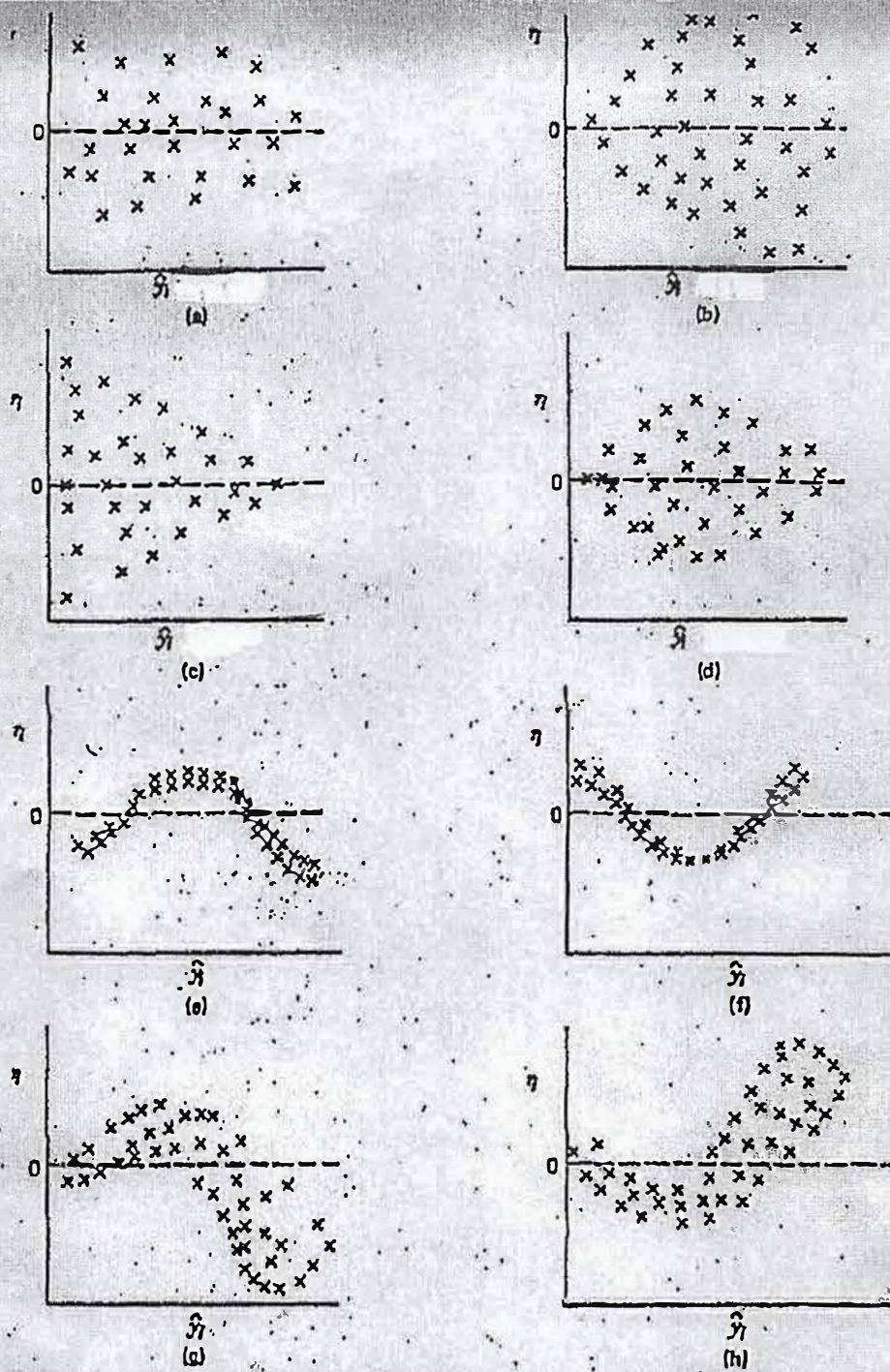


Figure 6.1 Residual plots: (a) null plot; (b) right-opening megaphone; (c) left-opening megaphone; (d) double outward bow; (e) nonlinearity; (f) nonlinearity; (g) nonlinearity and nonconstant variance; (h) nonlinearity and nonconstant variance